

CS 528: Unconditional Lower Bounds for Algorithms and Data Structures

Course Code	CS 528
Course Name	Unconditional Lower Bounds for Algorithms and Data Structures
L-T-P-C	3-0-0-3
Prerequisites	CS 208, CS 212 or equivalent
Intended for	M.Tech / Ph.D. / B.Tech 3rd and 4th year
Mutual Exclusion	None
Approval	64th BoA (May 2026)

Preamble

The theory of NP-completeness allows us to prove that a computational problem cannot have polynomial-time algorithms (i.e. a superpolynomial lower bound on the time required to solve the problem). However, such lower bounds rely on the unproven assumption that $P \neq NP$. In this course, we will study some techniques for proving unconditional lower bounds for many computational problems.

Course Modules

- **Unit 1: Motivation via streaming algorithms:** data stream model, estimation of frequency moments, connection with one-way communication. (5 hours)
- **Unit 2: One-way communication complexity:** definition of one-way protocols, definition of disjointness, index, and gap-hamming problem, lower bound for disjointness and index. (6 hours)
- **Unit 3: Lower bounds for streaming algorithms:** space lower bound for F_∞ estimation, reducing (one-way communication) gap-hamming problem to F_0 estimation, lower bound on the one-way communication complexity of gap-hamming. (8 hours)
- **Unit 4: Two-way communication complexity:** deterministic and randomized protocols, monochromatic rectangles, fooling sets, distribution complexity, corruption method. (12 hours)
- **Unit 5: Lower bounds for data structures:** the approximate nearest neighbour problem, cell-probe model, asymmetric communication complexity, lower bound for the approximate nearest neighbour problem. (11 hours)

Laboratory Modules

None

Text Books

1. Tim Roughgarden, *Communication Complexity (for Algorithm Designers)*, Now Publishers, 2016.

References

1. Amir Yehudayoff and Anup Rao, *Communication Complexity and Applications*, Cambridge University Press, 2020.
2. Eyal Kushilevitz and Noam Nisan, *Communication Complexity*, Cambridge University Press, 2009.